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Overlapping patch-based joint-sparse regression for hyperspectral image unmixing

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ABSTRACT

Sparse unmixing is thriving in the application of hyperspectral images (HSIs). Joint sparsity has been widely exploited to improve the abundance estimation accuracy. However, many previous joint sparsity based algorithms first decompose a 3-D HSI into a 2-D matrix, and the joint sparsity property is only considered along the horizontal and /or vertical directions. Therefore, the exploration of the local spatial information in HSIs is insufficient. In this article, we assume that pixels in a patch share the same set of endmembers from an available large spectral library, so the corresponding abundances of these pixels have similar sparse structures. Under this assumption, we design a new overlapping patch-based joint-sparse structure to enhance the mining of spatial information. Then, we propose an *overlapping patch-based joint-sparse and low-rank unmixing* (OPJLU) algorithm under the alternating direction method of multipliers (ADMM) framework to improve the local similarity utilization efficiency. We provide the residual and objective convergence results of the proposed algorithm. Experiments show the effectiveness of the proposed algorithm.

1. Introduction

With the development of remote sensing, hyperspectral images (HSIs) are accessible and widely used in many fields because of their rich spectral information [1–6]. Due to limitations in spatial resolution and impurities in the air, HSIs contain mixed pixels and noise, which hinders the further application of HSIs [7]. Thus, hyperspectral unmixing (HU) plays an important role in the application of HSIs, which decomposes each pixel in an HSI into several pure materials (called endmembers) and the corresponding fractions (termed abundances). Under the assumption of the linear mixture model (LMM), the mixed pixels are linearly mixed results of the endmembers [8].

When the endmember library is unknown, both the endmembers and the abundances need to be estimated. This process is referred to as blind unmixing. Assume that there are pure endmembers in pixels, some methods first extract the endmembers and then find their abundances, such as vertex component analysis (VCA) [9] for endmembers extraction and fully constrained least squares (FCLS) [10] for abundances estimation. However, in many practical scenarios, pure endmember assumption is hard to hold for all images, since some pixels are composed of highly mixed endmembers. Nonnegative matrix factorization (NMF) is broadly used to solve this problem [11]. It decomposes the unfolding matrix of the HSI into two smaller matrices, one of which represents the endmembers and the other matrix represents the corresponding abundances. Since the decomposition problem is nonconvex, there are many local minima, which will affect the accuracy of the decomposition. To improve the decomposition performance, many

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NMF-based algorithms are proposed and achieve promising results [12–17]. For example, most pixels contain a few endmembers, resulting in many zero values in abundances and indicating that the abundances are sparse. Consequently, some sparsity-constrained blind unmixing methods have been proposed for HU [12,16,18,19]. Additionally, the derivations of NMF, such as multilayer NMF and clustered multitask NMF, further enrich the application of NMF for HU [20,21]. To preserve the spatial information of HSIs, matrix–vector nonnegative tensor factorization (MV-NTF) is proposed for HU [22]. It decomposes the HSI into the sum of the outer product of the endmembers and the corresponding abundance maps. To further capture the spatial and spectral information of HSIs, many deep learning-based methods are introduced into HU and achieve excellent results [23–28].

With the growing availability and richness of spectral libraries, a new avenue emerges for addressing the HU problem. Pure spectrally constituent materials, endmembers, can be identified from potentially extensive spectral libraries and utilized to facilitate the unmixing process [29]. When the spectral libraries are known, we only need to estimate the abundances, which is regarded as nonblind unmixing. Typically, each pixel is composed of a few endmembers in the large spectral library. In other words, only a few elements in the abundance matrix are nonzero, that is, the abundance matrix is sparse. Therefore, the abundance estimation problem can be viewed as a sparse regression (SR) problem [30–35]. As the physical mechanism of abundances, various constraints are incorporated to improve the abundance estimation accuracy. To explore the sparsity of the abundance matrix, the l_p-norm is applied to the abundance matrix [16,36–40]. The *alternating direction sparse and low-rank unmixing* algorithm (ADSpLRU) introduces a weighted l_1-norm to enhance the sparsity of the abundance matrix and a weighted nuclear norm as a low-rank representation to describe the spatial correlation of pixels [31]. The *joint-sparse-blocks and low-rank unmixing* (JSpBLRU) incorporates the joint sparsity into HU by considering that the adjacent pixels share the same endmembers [33]. In addition, the *bilateral joint-sparse and low-rank unmixing* (BiJSpLRU) considers the joint sparsity in vertical and horizontal directions, offering more accurate abundance estimation results [32]. BiJSpLRU exploits the local joint sparsity of the abundance matrix from the vertical and horizontal directions separately, so each pixel is used twice. The *multidimensional low-rank representation* (MdLRR) for HU keeps the bilateral joint sparse structure in BiJSpLRU and further considers reshaping the abundance matrix into a 3-D abundance tensor and simultaneously imposing low-rank constraints on its different modes. This approach improves upon traditional methods that apply low-rank constraints solely on the abundance matrix, thereby maximizing the utilization of latent spatial information [41]. In addition, considering the neighboring pixels having similar fractional abundances for the same endmember, the *sparse unmixing via variable splitting augmented Lagrangian and total variation* (SUnSAL-TV) introduces a total variation (TV) regularizer to improve the accuracy of the results [42]. The above algorithms are implemented based on the alternating direction method of multipliers (ADMM) [43], which has been widely applied in signal processing and image processing [44]. Overall, these algorithms provide promising unmixing results. However, the exploration of local spatial information in HSIs is still insufficient. It is expected to further utilize spatial information to improve the unmixing performance.

In this article, we propose a new joint-sparse structure. In detail, pixels in a patch in HSIs are assumed to be similar, so the corresponding abundances have the same joint-sparse structures. Particularly, the overlapping patches come from a sliding square window with a size of 2×2 and a stride of one. Therefore, the spatial information of the HSIs is further mined. In summary, the contributions of this article are as follows:

1. We first propose a new joint-sparse structure that considers the corresponding abundances of pixels in a patch to have similar sparse structures. Then we design a new transformation matrix to realize this joint-sparse structure. In particular, each pixel in the overlapping patches is used multiple times, harnessing more spatial information from the HSIs.

2. We propose an *overlapping patch-based joint-sparse and low-rank unmixing* (OPJLU) algorithm, which is designed under the ADMM framework. We provide the residual and objective convergence results of the proposed algorithm. Experiment results show that OPJLU achieves better unmixing performance than some state-of-the-art algorithms.

The rest of the article is as follows. Section 2 reviews the related works on sparse regression problems. Section 3 proposes a new joint-sparse structure and a corresponding unmixing algorithm. Section 4 analyzes the convergence of OPJLU. Section 5 compares OPJLU and some state-of-the-art algorithms on simulated and real datasets.

2. Related works

In this section, we will introduce some basic concepts and models for HU. Given an HSI $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times l}$ with $n = n_1 \times n_2$ pixels and l spectral bands, and denote the mode-3 unfolding of $\mathcal{Y}$ as $\boldsymbol{Y} \in \mathbb{R}^{l \times n}$. According to the LMM, pixels in $\boldsymbol{Y}$ are considered as linear combinations of endmembers, i.e.,

$$\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{X} + \boldsymbol{E}, \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{l \times m}$ denotes an endmember matrix containing m spectral materials, $\boldsymbol{X} \in \mathbb{R}^{m \times n}$ is the unknown abundance matrix, and $\boldsymbol{E} \in \mathbb{R}^{l \times n}$ is the noise. Two physical factors are usually considered: the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC) [10]. The corresponding formulas are expressed as follows

$$\boldsymbol{X} \geq \boldsymbol{0},\ \boldsymbol{1}^T\boldsymbol{X} = \boldsymbol{1}^T, \tag{2}$$

where $\boldsymbol{1}$ is an array with all ones. Due to the variability of the endmembers, we only consider the ANC.

Since each pixel only contains a small number of endmembers in the spectral library, the sparsity of the abundance matrix is taken into account. Joint sparsity is an extension of the sparsity, which primarily considers the sparsity of the rows (or columns) of a matrix. The $l_{2,1}$-norm is usually introduced to characterize the joint sparsity, and it is defined as

$$\|\boldsymbol{X}\|_{2,1} = \sum_{i=1}^{m} \|\boldsymbol{X}^{[i]}\|_2, \tag{3}$$

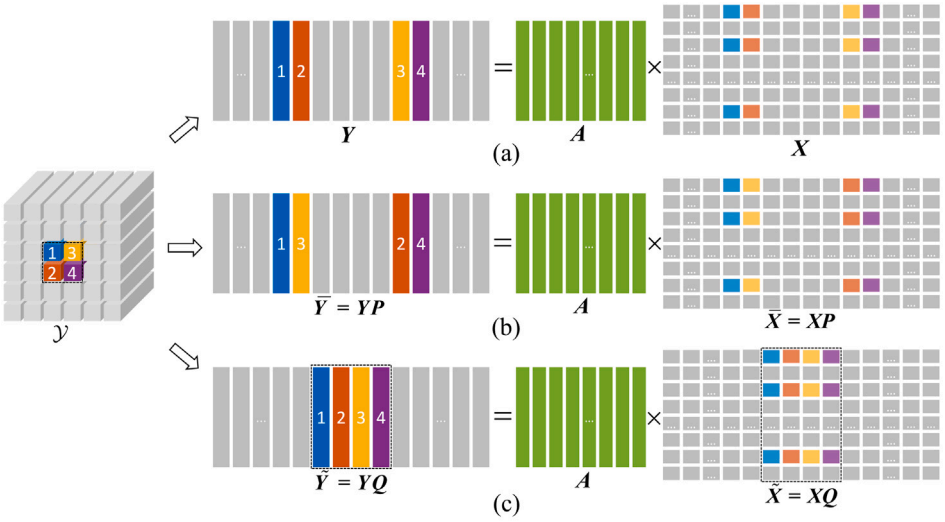


Fig. 1. Comparison of different expansion methods. (a) The expansion of the HSI $\mathcal{Y}$ along vertical direction. (b) The expansion of $\mathcal{Y}$ along horizontal direction. (c) The proposed expansion of $\mathcal{Y}$ by patches. Note that there is no need to construct the permutation matrix $\boldsymbol{P}$ and the transformation matrix $\boldsymbol{Q}$ explicitly.

where $\boldsymbol{X}^{[i]}$ is the ith row of $\boldsymbol{X}$. To define the joint-sparse-blocks structure, we first partition $\boldsymbol{X}$ as

$$\boldsymbol{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \dots, \boldsymbol{X}_s], \tag{4}$$

where $\boldsymbol{X}_j \in \mathbb{R}^{m\times d_j}$, for $j = 1, \dots, s$, $\sum_{j=1}^{s} d_j = n$, and then assume that each $\boldsymbol{X}_j$ is joint-sparse. Generally, a weighted $l_{2,1}$-norm is adopted for each block of $\boldsymbol{X}$ as

$$\|\boldsymbol{X}_j\|_{\boldsymbol{w}_j,2,1} = \sum_{i=1}^{m} w_{i,j}\|\boldsymbol{X}_j^{[i]}\|_2, \tag{5}$$

where $\boldsymbol{X}_j^{[i]} \in \mathbb{R}^{1\times d_j}$ is the ith row of the jth block of $\boldsymbol{X}$ and $\boldsymbol{w}_j = [w_{1,j}, w_{2,j}, \dots, w_{m,j}]$ is a nonnegative weight vector [33]. Based on this fact, JSpBLRU considers that pixels in a neighborhood contain the same endmembers, and exploits the joint-sparsity-blocks structure for HU. What is more, utilizing both the horizontal and vertical spatial information in HSIs, BiJSpLRU proposes a bilateral joint sparsity based model as

$$\begin{aligned} &\min_{\boldsymbol{X}} \ \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{A}\boldsymbol{X}\|_F^2 + \lambda_1 \sum_{j=1}^{s}\|\boldsymbol{X}_j\|_{\boldsymbol{W}_{1,j},2,1} + \lambda_2 \sum_{j=1}^{s}\|\bar{\boldsymbol{X}}_j\|_{\boldsymbol{W}_{2,j},2,1} + \tau\|\boldsymbol{X}\|_{\boldsymbol{W}_3,*} \\ &s.t.\ \boldsymbol{X} \geq \boldsymbol{0}, \end{aligned} \tag{6}$$

where $\|\boldsymbol{X}_j\|_{\boldsymbol{W}_{1,j},2,1} = \sum_{i=1}^{m}(\boldsymbol{W}_1)_{ij}\|\boldsymbol{X}_j^{[i]}\|_2$, $\|\bar{\boldsymbol{X}}_j\|_{\boldsymbol{W}_{2,j},2,1} = \sum_{i=1}^{m}(\boldsymbol{W}_2)_{ij}\|\bar{\boldsymbol{X}}_j^{[i]}\|_2$, $\|\boldsymbol{X}\|_{\boldsymbol{W}_3,*} = \sum_{i=1}^{m}(\boldsymbol{W}_3)_{ii}\sigma_i(\boldsymbol{X})$, λ_1, λ_2, and τ are regularization parameters, $\|\boldsymbol{X}\|_{\boldsymbol{W}_3,*}$ is the weighted nuclear norm of $\boldsymbol{X}$ [32,33,45–47], $\boldsymbol{X} = [\boldsymbol{X}_1, \dots, \boldsymbol{X}_s]$, $\bar{\boldsymbol{X}} = [\bar{\boldsymbol{X}}_1, \dots, \bar{\boldsymbol{X}}_s] = \boldsymbol{X}\boldsymbol{P} \in \mathbb{R}^{m\times n}$ is also an abundance matrix related to the horizontal expansion of the HSI, $\boldsymbol{P} \in \mathbb{R}^{n\times n}$ is a corresponding permutation matrix, $\boldsymbol{X}_j$ and $\bar{\boldsymbol{X}}_j \in \mathbb{R}^{m\times d_j}$ are the jth column blocks of $\boldsymbol{X}$ and $\bar{\boldsymbol{X}}$, respectively, $\sum_{j=1}^{s} d_j = n$, $\sigma_i(\boldsymbol{X})$ is the ith singular value of $\boldsymbol{X}$, and $\boldsymbol{W}_1 = [\boldsymbol{W}_{1,1}, \dots, \boldsymbol{W}_{1,s}] \in \mathbb{R}^{m\times s}$, $\boldsymbol{W}_2 = [\boldsymbol{W}_{2,1}, \dots, \boldsymbol{W}_{2,s}] \in \mathbb{R}^{m\times s}$, and $\boldsymbol{W}_3 = \mathbf{diag}(w_{3,1}, \dots, w_{3,m}) \in \mathbb{R}^{m\times m}$ are weighting matrices. It should be noted that the permutation matrix $\boldsymbol{P}$ is used to obtain another HSI matrix $\bar{\boldsymbol{Y}}$ by unfolding the HSI tensor $\mathcal{Y}$ in mode-3 along the horizontal direction, i.e., $\bar{\boldsymbol{Y}} = \boldsymbol{Y}\boldsymbol{P}$, and therefore $\bar{\boldsymbol{X}} = \boldsymbol{X}\boldsymbol{P}$. In particular, there is no need to construct $\boldsymbol{P}$ explicitly [32].

Additionally, block sparsity is a type of sparsity and has been widely applied to address various image processing and signal processing problems. A vector or a matrix with multiple blocks is called block sparse if only a few of its blocks have nonzero Euclidean norm or Frobenius norm [48]. There are various representations of block sparsity. For example, some representations use Markov modeling for block sparsity, and others employ hypothesis testing for block sparse recovery [49–51].

Notice that BiJSpLRU uses more spatial information to obtain more accurate abundance estimation. However, spatial information in the horizontal and vertical directions is separately considered, so the spatial information mining of BiJSpLRU is still inadequate. In the following, we mine more spatial information in HSIs to further improve the unmixing performance.

3. OPJLU

In this section, we first propose a new joint sparse structure. Then we give the proposed model and solve it under the ADMM framework.

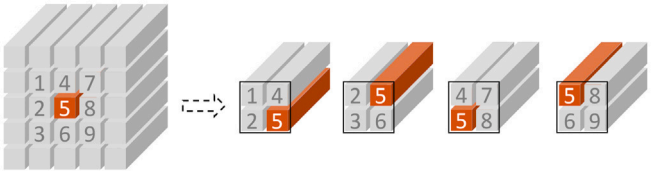


Fig. 2. Overlapping patches in OPJLU. For any pixel in the HSI, such as pixel #5, it is included in four overlapping and distinct patches.

3.1. Proposed overlapping patch-based joint-sparse structure

The joint sparsity of the abundance matrix has been well studied in the literature. Here, we propose to assume that pixels in a patch are composed of the same endmembers. For example, as shown in Fig. 1, the pixels in a patch are labeled as #1, #2, #3, and #4. JSpBLRU considers the joint sparsity of two blocks including pixels {#1, #2} and also {#3, #4}, and BiJSpLRU further considers the joint sparsity of two blocks including pixels {#1, #3} and also {#2, #4}, see Fig. 1 (a) and (b). Both of them do not consider the joint sparsity of block including pixels {#1, #2, #3, #4} together. Hence, leveraging spatial similarity, we assume that the marked pixels {#1, #2, #3, #4} have the same endmembers, as shown in Fig. 1 (c), so the corresponding abundances have the same sparse structure. That said, the abundances according to pixels {#1, #2, #3, #4} are joint sparse and therefore it is reasonable to cluster these pixels together and consider them simultaneously. In this vein, similarly considering all the pixels in $\boldsymbol{Y}$, we obtain a transformed HSI matrix $\tilde{\boldsymbol{Y}} \in \mathbb{R}^{l\times\tilde{n}}$ with

$$\tilde{\boldsymbol{Y}} = \boldsymbol{A}\tilde{\boldsymbol{X}}, \tag{7}$$

where $\tilde{\boldsymbol{X}} \in \mathbb{R}^{m\times\tilde{n}}$ is the corresponding transformed abundance matrix.

In the following, we show how to represent the matrix $\tilde{\boldsymbol{Y}}$ by introducing a *transformation matrix* $\boldsymbol{Q}$. To begin, we set the patches obtained from a sliding window with a size of 2×2 and a stride of one. It follows that each pixel is included in four overlapping patches, as shown in Fig. 2. Thus each pixel is utilized four times. Notably, we apply zero padding to the edges of the HSI $\mathcal{Y}$ when extracting patches. For simplicity, we remove these zero-padded pixels and obtain the transformed matrix $\tilde{\boldsymbol{Y}} \in \mathbb{R}^{l\times\tilde{n}}$ and $\tilde{n} = 4n$.

Next, by introducing a transformation matrix $\boldsymbol{Q}$, we have

$$\begin{aligned}\tilde{\boldsymbol{Y}} &= [\ldots, \boldsymbol{y}_i, \ldots, \boldsymbol{y}_i, \ldots, \boldsymbol{y}_i, \ldots, \boldsymbol{y}_i, \ldots]\\ &= \boldsymbol{Y}[\ldots, \boldsymbol{e}_i, \ldots, \boldsymbol{e}_i, \ldots, \boldsymbol{e}_i, \ldots, \boldsymbol{e}_i, \ldots]\\ &= \boldsymbol{Y}\boldsymbol{Q},\end{aligned} \tag{8}$$

where $\boldsymbol{y}_i$ is the ith column of $\boldsymbol{Y}$ and $\boldsymbol{e}_i$ is the ith column of an $n\times n$ identity matrix $\boldsymbol{I}$,

$$\boldsymbol{Q} = [\ldots, \boldsymbol{e}_i, \ldots, \boldsymbol{e}_i, \ldots, \boldsymbol{e}_i, \ldots, \boldsymbol{e}_i, \ldots] \in \mathbb{R}^{n\times 4n}. \tag{9}$$

It is easy to get

$$\boldsymbol{Q}\boldsymbol{Q}^T = \sum_{i=1}^{n} 4\boldsymbol{e}_i\boldsymbol{e}_i^T = 4\boldsymbol{I} \in \mathbb{R}^{n\times n}. \tag{10}$$

Clearly, from the above expansion, we also get the transformed abundance matrix $\tilde{\boldsymbol{X}} = \boldsymbol{X}\boldsymbol{Q}$. It is worth noting that there is no need to represent the matrix $\boldsymbol{Q}$ explicitly.

In this case, we propose to an overlapping patch-based joint-sparse on $\tilde{\boldsymbol{X}}$. Partition $\tilde{\boldsymbol{X}}$ as

$$\tilde{\boldsymbol{X}} = [\tilde{\boldsymbol{X}}_1, \tilde{\boldsymbol{X}}_2, \ldots, \tilde{\boldsymbol{X}}_t], \tag{11}$$

where $\tilde{\boldsymbol{X}}_j$ is the jth column block of $\tilde{\boldsymbol{X}}$, for $j = 1, \ldots, t$, and $t = (n_1 + 1)(n_2 + 1)$. We note that each column block $\tilde{\boldsymbol{X}}_j$ is obtained from a patch. Particularly, $\tilde{\boldsymbol{X}}_j \in \mathbb{R}^{m\times 4}$ if the corresponding patch contains only central pixels, $\tilde{\boldsymbol{X}}_j \in \mathbb{R}^{m\times 2}$ if the corresponding patch contains only boundary pixels, and $\tilde{\boldsymbol{X}}_j \in \mathbb{R}^{m\times 1}$ otherwise. Finally, we propose to consider the joint sparse on each block $\tilde{\boldsymbol{X}}_j$ and name it as the overlapping patch-based joint-sparse structure.

3.2. OPJLU

Combining the proposed joint sparse structure of $\tilde{\boldsymbol{X}}$, via the $l_{2,1}$-norm on each block $\tilde{\boldsymbol{X}}_j$, and the low-rank constraint of $\boldsymbol{X}$, via the weighted nuclear norm, the proposed model is as follows

$$\begin{aligned}&\min_{\boldsymbol{X}} \ \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{A}\boldsymbol{X}\|_F^2 + \lambda\sum_{j=1}^{t}\|\tilde{\boldsymbol{X}}_j\|_{\boldsymbol{\Omega}_1,2,1} + \tau\|\boldsymbol{X}\|_{\boldsymbol{\Omega}_2,*}\\ &s.t.\ \boldsymbol{X} \geq \boldsymbol{0},\ \tilde{\boldsymbol{X}} = \boldsymbol{X}\boldsymbol{Q},\end{aligned} \tag{12}$$

Algorithm 1: Pseudocode of the OPJLU Algorithm.

Input: $\boldsymbol{Y}$, $\boldsymbol{A}$;
Select parameters: λ, τ, μ and maximum iterations.
Initialization: $\boldsymbol{V}^0$, $\boldsymbol{\Lambda}^0$ and set $k = 0$.
repeat
  Update $\boldsymbol{X}^{k+1}$ by (21).
  Update $\boldsymbol{\Omega}_1^{k+1}$ by (22) and $\boldsymbol{\Omega}_2^{k+1}$ by (23).
  Update $\boldsymbol{V}_1^{k+1}$ by (25), $\boldsymbol{V}_2^{k+1}$ by (26), and $\boldsymbol{V}_3^{k+1}$ by (27).
  Update $\boldsymbol{\Lambda}^{k+1}$ by (28).
until some stopping criterion is satisfied;
Output: $\hat{\boldsymbol{X}} = \boldsymbol{X}^{k+1}$.

where $\boldsymbol{\Omega}_1$ and $\boldsymbol{\Omega}_2$ are nonnegative weighting matrices, which will be automatically updated later. After introducing three variables, the proposed model becomes

$$\begin{aligned} &\min_{\boldsymbol{X},\boldsymbol{V}_1,\boldsymbol{V}_2,\boldsymbol{V}_3} \ \frac{1}{2}\|\boldsymbol{Y}-\boldsymbol{A}\boldsymbol{X}\|_F^2 + \lambda\sum_{j=1}^{t}\|\boldsymbol{V}_{1,j}\|_{\boldsymbol{\Omega}_1,2,1} + \tau\|\boldsymbol{V}_2\|_{\boldsymbol{\Omega}_2,*} + \iota_{\mathbb{R}^+}(\boldsymbol{V}_3) \\ &s.t.\ \boldsymbol{X}\boldsymbol{Q} = \boldsymbol{V}_1,\ \boldsymbol{X} = \boldsymbol{V}_2,\ \boldsymbol{X} = \boldsymbol{V}_3, \end{aligned} \tag{13}$$

where $\iota_{\mathbb{R}^+}(v)$ is the indicator function, meaning $\iota_{\mathbb{R}^+}(v) = 0$ if $v \geq 0$ and $\iota_{\mathbb{R}^+}(v) = +\infty$ otherwise. For convenience, we denote

$$\begin{aligned} g_1(\boldsymbol{X}) &= \frac{1}{2}\|\boldsymbol{Y}-\boldsymbol{A}\boldsymbol{X}\|_F^2, \\ g_2(\boldsymbol{V},\boldsymbol{\Omega}) &= \lambda\sum_{j=1}^{t}\|\boldsymbol{V}_{1,j}\|_{\boldsymbol{\Omega}_1,2,1} + \tau\|\boldsymbol{V}_2\|_{\boldsymbol{\Omega}_2,*} + \iota_{\mathbb{R}^+}(\boldsymbol{V}_3), \end{aligned} \tag{14}$$

where we define $\boldsymbol{V} = (\boldsymbol{V}_1, \boldsymbol{V}_2, \boldsymbol{V}_3) \in \mathbb{R}^{m\times 6n}$, $\boldsymbol{H} = (\boldsymbol{Q}, \boldsymbol{I}, \boldsymbol{I}) \in \mathbb{R}^{n\times 6n}$, and $\boldsymbol{\Omega} = (\boldsymbol{\Omega}_1, \boldsymbol{\Omega}_2) \in \mathbb{R}^{m\times(t+m)}$, and $\boldsymbol{I}$ is an n-by-n identity matrix. Then, we rewrite (13) as

$$\begin{aligned} &\min_{\boldsymbol{X},\boldsymbol{V}} \ g_1(\boldsymbol{X}) + g_2(\boldsymbol{V},\boldsymbol{\Omega}) \\ &s.t.\ \boldsymbol{X}\boldsymbol{H} = \boldsymbol{V}. \end{aligned} \tag{15}$$

Define the augmented Lagrangian function

$$\mathcal{L}_\mu(\boldsymbol{X},\boldsymbol{V},\boldsymbol{\Omega},\boldsymbol{\Lambda}) = g_1(\boldsymbol{X}) + g_2(\boldsymbol{V},\boldsymbol{\Omega}) + \langle\boldsymbol{\Lambda}, \boldsymbol{X}\boldsymbol{H}-\boldsymbol{V}\rangle + \frac{\mu}{2}\|\boldsymbol{X}\boldsymbol{H}-\boldsymbol{V}\|_F^2, \tag{16}$$

where $\mu > 0$ is a penalty parameter, $\boldsymbol{\Lambda} = (\boldsymbol{\Lambda}_1, \boldsymbol{\Lambda}_2, \boldsymbol{\Lambda}_3) \in \mathbb{R}^{m\times 6n}$ is the Lagrange multiplier, and $\langle\boldsymbol{\Lambda}, \boldsymbol{X}\boldsymbol{H}-\boldsymbol{V}\rangle = \mathrm{Tr}(\boldsymbol{\Lambda}^T(\boldsymbol{X}\boldsymbol{H}-\boldsymbol{V}))$ is the inner product of two matrices. Then the iterative scheme is proposed as follows:

$$\begin{cases} \boldsymbol{X}^{k+1} = \arg\min_{\boldsymbol{X}} \ \mathcal{L}_\mu(\boldsymbol{X}, \boldsymbol{V}^k, \boldsymbol{\Omega}^k, \boldsymbol{\Lambda}^k) \\ \text{update } \boldsymbol{\Omega}^{k+1} \\ \boldsymbol{V}^{k+1} = \arg\min_{\boldsymbol{V}} \ \mathcal{L}_\mu(\boldsymbol{X}^{k+1}, \boldsymbol{V}, \boldsymbol{\Omega}^{k+1}, \boldsymbol{\Lambda}^k) \\ \boldsymbol{\Lambda}^{k+1} = \boldsymbol{\Lambda}^k + \mu(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^{k+1}). \end{cases} \tag{17}$$

We note that $\boldsymbol{\Omega}^{k+1}$ will be automatically updated, similarly as in [33]. Additionally, two operators in the following will be used to solve the subproblems:

(1) The vect-soft operator:

$$\textbf{vect-soft}_\alpha(\boldsymbol{x}) = \boldsymbol{x}\frac{\max\{\|\boldsymbol{x}\|_2 - \alpha, 0\}}{\max\{\|\boldsymbol{x}\|_2 - \alpha, 0\} + \alpha}, \tag{18}$$

for $\boldsymbol{x} \in \mathbb{R}^n$ and $\alpha > 0$. The operator is a vectorial soft-thresholding operator, and it acts on the norm of the entire vector and is suitable for sparsity problems that consider the overall structure [52].

(2) The weighted singular value thresholding operator (SVT):

$$\textbf{SVT}_{\mathbf{S},\beta}(\boldsymbol{X}) = \tilde{\boldsymbol{U}}\,\textbf{diag}(\max(\tilde{\sigma_1} - \beta s_{11}, 0), \ldots, \max(\tilde{\sigma_r} - \beta s_{rr}, 0))\tilde{\boldsymbol{V}}^T, \tag{19}$$

where $\tilde{\boldsymbol{U}}$, $\tilde{\boldsymbol{V}}$, and $\tilde{\boldsymbol{\Sigma}} = \textbf{diag}(\tilde{\sigma_1}, \ldots, \tilde{\sigma_r})$ come from the singular value decomposition (SVD) of $\boldsymbol{X}$, i.e., $\boldsymbol{X} = \tilde{\boldsymbol{U}}\tilde{\boldsymbol{\Sigma}}\tilde{\boldsymbol{V}}^T$, $\boldsymbol{S} = \textbf{diag}(s_{11}, \ldots, s_{rr})$ is a nonnegative weighting diagonal matrix, and β is a positive parameter.

In the following, we give detailed analysis of each subproblem. For $\boldsymbol{X}$-subproblem, we have

$$\boldsymbol{X}^{k+1} = \arg\min_{\boldsymbol{X}} \ g_1(\boldsymbol{X}) + \langle\boldsymbol{\Lambda}^k, \boldsymbol{X}\boldsymbol{H}-\boldsymbol{V}^k\rangle + \frac{\mu}{2}\|\boldsymbol{X}\boldsymbol{H}-\boldsymbol{V}^k\|_F^2. \tag{20}$$

Recall that $\boldsymbol{Q}$ satisfies $\boldsymbol{Q}\boldsymbol{Q}^T = 4\boldsymbol{I}$, so $\boldsymbol{H}$ satisfies $\boldsymbol{H}\boldsymbol{H}^T = 6\boldsymbol{I}$, then we get

$$\boldsymbol{X}^{k+1} = (\boldsymbol{A}^T\boldsymbol{A} + 6\mu\hat{\boldsymbol{I}})^{-1}(\boldsymbol{A}^T\boldsymbol{Y} + (\mu\boldsymbol{V}^k - \boldsymbol{\Lambda}^k)\boldsymbol{H}^T), \tag{21}$$

where $\hat{\boldsymbol{I}}$ is an m-by-m identity matrix. After obtaining $\boldsymbol{X}^{k+1}$, the next step is to update the two weights $\boldsymbol{\Omega}_1$ and $\boldsymbol{\Omega}_2$. Similarly as in [32], the updates for $\boldsymbol{\Omega}_1$ and $\boldsymbol{\Omega}_2$ are

$$(\boldsymbol{\Omega}_1^{k+1})_{ij} = \frac{1}{\|((\boldsymbol{X}^{k+1}\boldsymbol{Q})_j + \frac{1}{\mu}\boldsymbol{\Lambda}_{1,j}^k)^{[i]}\|_2 + \varepsilon}, \tag{22}$$

$$(\boldsymbol{\Omega}_2^{k+1})_{ii} = \frac{1}{\sigma_i(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_2^k) + \varepsilon}, \tag{23}$$

for $i = 1, \dots, m$, and $j = 1, \dots, t$, where ε is a small constant used to avoid singularities and $\sigma_i(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_2^k)$ is the ith singular value of $\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_2^k$. These will be used in the following $\boldsymbol{V}$-subproblem. The weighting coefficients in $\boldsymbol{\Omega}_1$ are used to enhance the row sparsity of $\boldsymbol{V}_1$, and the weighting coefficients in $\boldsymbol{\Omega}_2$ aim to enhance the sparsity of singular values of $\boldsymbol{V}_2$. The $\boldsymbol{V}_1$-subproblem is

$$\boldsymbol{V}_1^{k+1} = \arg\min_{\boldsymbol{V}_1} \lambda\sum_{j=1}^{t}\|\boldsymbol{V}_{1,j}\|_{\boldsymbol{\Omega}_{1,j}^{k+1},2,1} - \langle\boldsymbol{\Lambda}_1^k, \boldsymbol{V}_1\rangle + \frac{\mu}{2}\|\boldsymbol{X}^{k+1}\boldsymbol{Q} - \boldsymbol{V}_1\|_F^2. \tag{24}$$

Similarly as in [32], we get that the solution of the ith row of the jth column block of $\boldsymbol{V}_1$ is

$$(\boldsymbol{V}_{1,j}^{k+1})^{[i]} = \textbf{vect-soft}_{\frac{\lambda}{\mu}(\boldsymbol{\Omega}_1^{k+1})_{ij}}(((\boldsymbol{X}^{k+1}\boldsymbol{Q})_j + \frac{1}{\mu}\boldsymbol{\Lambda}_{1,j}^k)^{[i]}), \tag{25}$$

for $i = 1, \dots, m$, and $j = 1, \dots, t$. Considering the $\boldsymbol{V}_2$ and $\boldsymbol{V}_3$-subproblems, it is easy to have that

$$\begin{aligned}\boldsymbol{V}_2^{k+1} &= \arg\min_{\boldsymbol{V}_2} \tau\|\boldsymbol{V}_2\|_{\boldsymbol{\Omega}_2^{k+1},*} - \langle\boldsymbol{\Lambda}_2^k, \boldsymbol{V}_2\rangle + \frac{\mu}{2}\|\boldsymbol{X}^{k+1} - \boldsymbol{V}_2\|_F^2 \\ &= \textbf{SVT}_{\boldsymbol{\Omega}_2^{k+1},\frac{\tau}{\mu}}(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_2^k),\end{aligned} \tag{26}$$

$$\begin{aligned}\boldsymbol{V}_3^{k+1} &= \arg\min_{\boldsymbol{V}_3} \iota_{\mathbb{R}^+}(\boldsymbol{V}_3) - \langle\boldsymbol{\Lambda}_3^k, \boldsymbol{V}_3\rangle + \frac{\mu}{2}\|\boldsymbol{X}^{k+1} - \boldsymbol{V}_3\|_F^2 \\ &= \max(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_3^k, \boldsymbol{0}),\end{aligned} \tag{27}$$

and update the multipliers as

$$\begin{cases}\boldsymbol{\Lambda}_1^{k+1} = \boldsymbol{\Lambda}_1^k + \mu(\boldsymbol{X}^{k+1}\boldsymbol{Q} - \boldsymbol{V}_1^{k+1}) \\ \boldsymbol{\Lambda}_2^{k+1} = \boldsymbol{\Lambda}_2^k + \mu(\boldsymbol{X}^{k+1} - \boldsymbol{V}_2^{k+1}) \\ \boldsymbol{\Lambda}_3^{k+1} = \boldsymbol{\Lambda}_3^k + \mu(\boldsymbol{X}^{k+1} - \boldsymbol{V}_3^{k+1}).\end{cases} \tag{28}$$

We summarize the proposed algorithm named OPJLU in Algorithm 1.

Finally, regarding the computational complexity of OPJLU, we note that the most expensive step is the **SVT** operator in the update of $\boldsymbol{V}_2$, which has the computational complexity of $O(mn\min(m,n))$. The complexity of other steps is $O(m^2n)$. Here, we recall that m denotes the number of spectral materials and n is the number of pixels. Generally, we have $n > m$, and therefore, the overall computational complexity of OPJLU is $O(m^2n)$.

4. Convergence analysis

To obtain the convergence analysis of the proposed algorithm, we first give the following assumption and three lemmas.

Assumption 1. The unaugmented Lagrangian $\mathcal{L}_0(\boldsymbol{X}, \boldsymbol{V}, \boldsymbol{\Omega}, \boldsymbol{\Lambda})$ has a saddle point. Therefore, there exists a point $(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star, \boldsymbol{\Lambda}^\star)$ that satisfies

$$\mathcal{L}_0(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star, \boldsymbol{\Lambda}) \le \mathcal{L}_0(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star, \boldsymbol{\Lambda}^\star) \le \mathcal{L}_0(\boldsymbol{X}, \boldsymbol{V}, \boldsymbol{\Omega}, \boldsymbol{\Lambda}^\star), \tag{29}$$

which is suitable for all $\boldsymbol{X}$, $\boldsymbol{V}$, $\boldsymbol{\Omega}$, $\boldsymbol{\Lambda}$.

From Assumption 1, $\mathcal{L}_0(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star, \boldsymbol{\Lambda}^\star)$ is finite and $(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star)$ is a solution to (15). Let $(\boldsymbol{X}^k, \boldsymbol{V}^k, \boldsymbol{\Omega}^k, \boldsymbol{\Lambda}^k)$ be the sequence generated by (17). We have $\boldsymbol{X}^\star\boldsymbol{H} - \boldsymbol{V}^\star = \boldsymbol{0}$, $g_1(\boldsymbol{X}^\star) < \infty$, and $g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^\star) < \infty$. Define

$$\begin{aligned}p^k &= g_1(\boldsymbol{X}^k) + g_2(\boldsymbol{V}^k, \boldsymbol{\Omega}^k), \\ \boldsymbol{r}^k &= \boldsymbol{X}^k\boldsymbol{H} - \boldsymbol{V}^k, \\ p^\star &= g_1(\boldsymbol{X}^\star) + g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^\star).\end{aligned} \tag{30}$$

Lemma 1. From *Assumption* 1, we obtain

$$-\langle\boldsymbol{\Lambda}^\star, \mathbf{r}^{k+1}\rangle \le p^{k+1} - p^\star. \tag{31}$$

Proof. Since $(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star, \boldsymbol{\Lambda}^\star)$ is a saddle point, we have

$$\mathcal{L}_0(\boldsymbol{X}^\star, \boldsymbol{V}^\star, \boldsymbol{\Omega}^\star, \boldsymbol{\Lambda}^\star) \leq \mathcal{L}_0(\boldsymbol{X}^{k+1}, \boldsymbol{V}^{k+1}, \boldsymbol{\Omega}^{k+1}, \boldsymbol{\Lambda}^\star), \tag{32}$$

which means

$$\begin{aligned} &g_1(\boldsymbol{X}^\star) + g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^\star) + \langle \boldsymbol{\Lambda}^\star, \boldsymbol{X}^\star \boldsymbol{H} - \boldsymbol{V}^\star \rangle \\ &\leq g_1(\boldsymbol{X}^{k+1}) + g_2(\boldsymbol{V}^{k+1}, \boldsymbol{\Omega}^{k+1}) + \langle \boldsymbol{\Lambda}^\star, \boldsymbol{X}^{k+1} \boldsymbol{H} - \boldsymbol{V}^{k+1} \rangle. \end{aligned} \tag{33}$$

Combining $\boldsymbol{X}^\star \boldsymbol{H} - \boldsymbol{V}^\star = \boldsymbol{0}$ and (33), we get $p^\star \leq p^{k+1} - \langle \boldsymbol{\Lambda}^\star, \boldsymbol{r}^{k+1} \rangle$. □

Lemma 2. From *Lemma* 1, if $\boldsymbol{\Omega}^{k+1} \leq \boldsymbol{\Omega}^\star$, then we have

$$p^{k+1} - p^\star \leq -\langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} - (\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star) \rangle. \tag{34}$$

Proof. Recall that $\boldsymbol{X}^{k+1}$ minimizes $\mathcal{L}_\mu(\boldsymbol{X}, \boldsymbol{V}^k, \boldsymbol{\Omega}^k, \boldsymbol{\Lambda}^k)$, which minimizes

$$g_1(\boldsymbol{X}) + \langle \boldsymbol{\Lambda}^k, \boldsymbol{X}\boldsymbol{H} - \boldsymbol{V}^k \rangle + \frac{\mu}{2} \|\boldsymbol{X}\boldsymbol{H} - \boldsymbol{V}^k\|_F^2. \tag{35}$$

Since g_1 is differentiable, the optimality condition is

$$\boldsymbol{0} = \nabla g_1(\boldsymbol{X}^{k+1}) + \boldsymbol{\Lambda}^k \boldsymbol{H}^T + \mu(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^k)\boldsymbol{H}^T. \tag{36}$$

Recall that $\boldsymbol{\Lambda}^{k+1} = \boldsymbol{\Lambda}^k + \mu(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^{k+1})$, we have

$$\begin{aligned} \boldsymbol{0} &= \nabla g_1(\boldsymbol{X}^{k+1}) + (\boldsymbol{\Lambda}^{k+1} - \mu(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^{k+1}))\boldsymbol{H}^T + \mu(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^k)\boldsymbol{H}^T \\ &= \nabla g_1(\boldsymbol{X}^{k+1}) + (\boldsymbol{\Lambda}^{k+1} + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{V}^k))\boldsymbol{H}^T. \end{aligned} \tag{37}$$

It implies that $\boldsymbol{X}^{k+1}$ minimizes

$$g_1(\boldsymbol{X}) + \langle \boldsymbol{\Lambda}^{k+1} + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{V}^k), \boldsymbol{X}\boldsymbol{H} \rangle. \tag{38}$$

So, we get

$$\begin{aligned} &g_1(\boldsymbol{X}^{k+1}) + \langle \boldsymbol{\Lambda}^{k+1} + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{V}^k), \boldsymbol{X}^{k+1}\boldsymbol{H} \rangle \\ &\leq g_1(\boldsymbol{X}^\star) + \langle \boldsymbol{\Lambda}^{k+1} + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{V}^k), \boldsymbol{X}^\star \boldsymbol{H} \rangle. \end{aligned} \tag{39}$$

Similarly, $\boldsymbol{V}^{k+1}$ minimizes $\mathcal{L}_\mu(\boldsymbol{X}^{k+1}, \boldsymbol{V}, \boldsymbol{\Omega}^{k+1}, \boldsymbol{\Lambda}^k)$. Let

$$\bar{g}_2(\boldsymbol{V}) = g_2(\boldsymbol{V}, \boldsymbol{\Omega}^{k+1}), \tag{40}$$

so that $\boldsymbol{V}^{k+1}$ equivalently minimizes

$$\bar{g}_2(\boldsymbol{V}) + \langle \boldsymbol{\Lambda}^k, \boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V} \rangle + \frac{\mu}{2} \|\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}\|_F^2. \tag{41}$$

Since $\bar{g}_2$ is subdifferentiable, the optimality condition is

$$\boldsymbol{0} = \partial \bar{g}_2(\boldsymbol{V}^{k+1}) - \boldsymbol{\Lambda}^k + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{X}^{k+1}\boldsymbol{H}). \tag{42}$$

Recall that $\boldsymbol{\Lambda}^{k+1} = \boldsymbol{\Lambda}^k + \mu(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^{k+1})$, we have

$$\begin{aligned} \boldsymbol{0} &= \partial \bar{g}_2(\boldsymbol{V}^{k+1}) - \boldsymbol{\Lambda}^k + (\boldsymbol{\Lambda}^k - \boldsymbol{\Lambda}^{k+1}) \\ &= \partial \bar{g}_2(\boldsymbol{V}^{k+1}) - \boldsymbol{\Lambda}^{k+1}. \end{aligned} \tag{43}$$

This implies that $\boldsymbol{V}^{k+1}$ minimizes

$$\bar{g}_2(\boldsymbol{V}) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V} \rangle = g_2(\boldsymbol{V}, \boldsymbol{\Omega}^{k+1}) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V} \rangle. \tag{44}$$

Then, we have

$$g_2(\boldsymbol{V}^{k+1}, \boldsymbol{\Omega}^{k+1}) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V}^{k+1} \rangle \leq g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^{k+1}) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V}^\star \rangle. \tag{45}$$

Under the assumption of $\boldsymbol{\Omega}^{k+1} \leq \boldsymbol{\Omega}^\star$ and from the definition of g_2, we have

$$g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^{k+1}) \leq g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^\star). \tag{46}$$

Combining (45) and (46), we have

$$g_2(\boldsymbol{V}^{k+1}, \boldsymbol{\Omega}^{k+1}) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V}^{k+1} \rangle \leq g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^\star) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V}^\star \rangle. \tag{47}$$

Then (39) and (47) gives

$$\begin{aligned} &g_1(\boldsymbol{X}^{k+1}) + \langle \boldsymbol{\Lambda}^{k+1} + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{V}^k), \boldsymbol{X}^{k+1}\boldsymbol{H} \rangle + g_2(\boldsymbol{V}^{k+1}, \boldsymbol{\Omega}^{k+1}) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V}^{k+1} \rangle \\ &\leq g_1(\boldsymbol{X}^\star) + \langle \boldsymbol{\Lambda}^{k+1} + \mu(\boldsymbol{V}^{k+1} - \boldsymbol{V}^k), \boldsymbol{X}^\star \boldsymbol{H} \rangle + g_2(\boldsymbol{V}^\star, \boldsymbol{\Omega}^\star) - \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{V}^\star \rangle. \end{aligned} \tag{48}$$

Simplifying, we get

$$p^{k+1} + \langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \mu \, \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, \boldsymbol{X}^{k+1}\boldsymbol{H} \rangle \leq p^\star + \mu \, \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, \boldsymbol{X}^\star \boldsymbol{H} \rangle. \tag{49}$$

After moving and merging items, we get

$$\begin{aligned} p^{k+1} - p^\star &\leq -\langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, (\boldsymbol{X}^\star - \boldsymbol{X}^{k+1})\boldsymbol{H} \rangle \\ &= -\langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -(\boldsymbol{X}^{k+1}\boldsymbol{H} - \boldsymbol{V}^{k+1}) - \boldsymbol{V}^{k+1} + \boldsymbol{V}^\star \rangle \\ &= -\langle \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} - (\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star) \rangle. \end{aligned} \tag{50}$$

□

Lemma 3. Let

$$\phi^k = \frac{1}{\mu}\|\boldsymbol{\Lambda}^k - \boldsymbol{\Lambda}^\star\|_F^2 + \mu \|\boldsymbol{V}^k - \boldsymbol{V}^\star\|_F^2. \tag{51}$$

Under *Lemmas* 1 *and* 2, we have

$$\phi^{k+1} - \phi^k \leq -\mu \|\boldsymbol{r}^{k+1} - (\boldsymbol{V}^{k+1} - \boldsymbol{V}^k)\|_F^2. \tag{52}$$

Proof. Combining (31) and (34), we have

$$0 \leq \langle \boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle + \mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} - (\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star) \rangle. \tag{53}$$

Let

$$\begin{aligned} \psi_1 &= 2\langle \boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle, \\ \psi_2 &= 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} - (\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star) \rangle. \end{aligned} \tag{54}$$

It is easy to get $\psi_1 + \psi_2 \geq 0$.

For ψ_1, since $\boldsymbol{\Lambda}^{k+1} = \boldsymbol{\Lambda}^k + \mu \boldsymbol{r}^{k+1}$ and $\boldsymbol{r}^{k+1} = \frac{1}{\mu}(\boldsymbol{\Lambda}^{k+1} - \boldsymbol{\Lambda}^k)$, we have

$$\begin{aligned} \psi_1 &= 2\langle \boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^{k+1}, \boldsymbol{r}^{k+1} \rangle \\ &= 2\langle \boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^k - \mu \boldsymbol{r}^{k+1}, \boldsymbol{r}^{k+1} \rangle \\ &= 2\langle \boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^k, \boldsymbol{r}^{k+1} \rangle - 2\mu \|\boldsymbol{r}^{k+1}\|_F^2 \\ &= \frac{2}{\mu}\langle \boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^k, \boldsymbol{\Lambda}^{k+1} - \boldsymbol{\Lambda}^k \rangle - \frac{1}{\mu}\|\boldsymbol{\Lambda}^{k+1} - \boldsymbol{\Lambda}^k\|_F^2 - \mu \|\boldsymbol{r}^{k+1}\|_F^2 \\ &= -\frac{1}{\mu}\|\boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^{k+1}\|_F^2 + \frac{1}{\mu}\|\boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^k\|_F^2 - \mu \|\boldsymbol{r}^{k+1}\|_F^2. \end{aligned} \tag{55}$$

For ψ_2, we have

$$\begin{aligned} \psi_2 &= 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} - (\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star) \rangle \\ &= 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} \rangle - 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, \boldsymbol{V}^{k+1} - \boldsymbol{V}^\star \rangle \\ &= 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} \rangle - 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, \boldsymbol{V}^{k+1} - \boldsymbol{V}^k - \boldsymbol{V}^\star + \boldsymbol{V}^k \rangle \\ &= 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, -\boldsymbol{r}^{k+1} \rangle - 2\mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k\|_F^2 - 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, \boldsymbol{V}^k - \boldsymbol{V}^\star \rangle \\ &= -\mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k + \boldsymbol{r}^{k+1}\|_F^2 - \mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k\|_F^2 - 2\mu \langle \boldsymbol{V}^{k+1} - \boldsymbol{V}^k, \boldsymbol{V}^k - \boldsymbol{V}^\star \rangle + \mu \|\boldsymbol{r}^{k+1}\|_F^2 \\ &= -\mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k + \boldsymbol{r}^{k+1}\|_F^2 - \mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star\|_F^2 + \mu \|\boldsymbol{V}^k - \boldsymbol{V}^\star\|_F^2 + \mu \|\boldsymbol{r}^{k+1}\|_F^2. \end{aligned} \tag{56}$$

Thus, (55) and (56) give

$$\begin{aligned} \psi_1 + \psi_2 &= -\frac{1}{\mu}\|\boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^{k+1}\|_F^2 + \frac{1}{\mu}\|\boldsymbol{\Lambda}^\star - \boldsymbol{\Lambda}^k\|_F^2 - \mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k + \boldsymbol{r}^{k+1}\|_F^2 - \mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^\star\|_F^2 + \mu \|\boldsymbol{V}^k - \boldsymbol{V}^\star\|_F^2 \\ &= -\phi^{k+1} + \phi^k - \mu \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k + \boldsymbol{r}^{k+1}\|_F^2 \\ &\geq 0. \end{aligned} \tag{57}$$

□

Then, we get the convergence result of Algorithm 1.

Theorem 1. *Under Assumption* 1, *let* $\boldsymbol{\Omega}^{k+1} \leq \boldsymbol{\Omega}^\star$ *and* $\{\boldsymbol{V}^k\}$ *converge, then the iterates* $(\boldsymbol{X}^k, \boldsymbol{V}^k, \boldsymbol{\Omega}^k, \boldsymbol{\Lambda}^k)$ *satisfy the following:*

(1) Residual convergence: $\boldsymbol{r}^k \to \boldsymbol{0}$ *as* $k \to \infty$.
(2) Objective convergence: $p^k \to p^\star$ *as* $k \to \infty$.

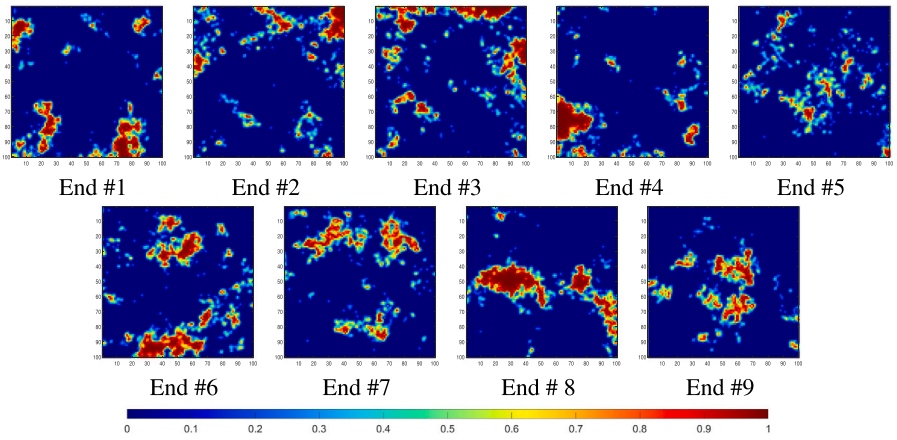


Fig. 3. Reference abundance maps of selected endmembers for Dataset 1.

Table 1
SRE (dB) and RMSE of different algorithms on Datasets 1 and 2.

Dataset 1						
Algorithms	SNR = 20 dB		SNR = 30 dB		SNR = 40 dB	
	SRE	RMSE	SRE	RMSE	SRE	RMSE
SUnSAL-TV	7.85	0.0293	12.99	0.0162	19.97	0.0073
ADSpLRU	4.51	0.0431	17.13	0.0101	25.32	0.0039
JSpBLRU	9.98	0.0230	18.82	0.0083	28.44	0.0027
BiJSpLRU	12.12	0.0180	20.43	0.0069	29.71	0.0024
MdLRR	**14.08**	**0.0143**	21.13	0.0064	29.15	0.0025
OPJLU	13.54	0.0152	**21.99**	**0.0058**	**31.47**	**0.0019**
Dataset 2						
Algorithms	SNR = 20 dB		SNR = 30 dB		SNR = 40 dB	
	SRE	RMSE	SRE	RMSE	SRE	RMSE
SUnSAL-TV	6.57	0.0217	12.81	0.0106	22.11	0.0036
ADSpLRU	5.44	0.0247	16.74	0.0067	22.81	0.0033
JSpBLRU	7.02	0.0206	18.57	0.0054	28.12	0.0018
BiJSpLRU	11.19	0.0127	18.26	0.0056	28.25	0.0018
MdLRR	10.26	0.0142	15.31	0.0079	21.78	0.0038
OPJLU	**12.55**	**0.0109**	**18.59**	**0.0054**	**29.16**	**0.0016**

Proof.

(1) Lemma 3 shows that $\{\phi^k\}$ is monotonically decreasing and $\phi^k \geq 0$, which indicates $\{\phi^k\}$ converges. Combined that $\{\boldsymbol{V}^k\}$ converges, we obtain that $\{\boldsymbol{\Lambda}^k\}$ converges. For (52), summed with respect to k, we obtain

$$\phi^0 \geq \mu \sum_{k=0}^{\infty} \|\boldsymbol{V}^{k+1} - \boldsymbol{V}^k + \boldsymbol{r}^{k+1}\|_F^2.$$

This indicates $\boldsymbol{V}^{k+1} - \boldsymbol{V}^k + \boldsymbol{r}^{k+1} \to \boldsymbol{0}$ as $k \to \infty$. Because of the convergence of $\{\boldsymbol{V}^k\}$, we get $\boldsymbol{r}^{k+1} \to \boldsymbol{0}$ as $k \to \infty$. Then we have that $\{\boldsymbol{X}^k\}$ converges.

(2) Recall that $\{\boldsymbol{V}^k\}$ converges and $\boldsymbol{r}^{k+1} \to \boldsymbol{0}$, so combining (31) and (34) gives $p^k \to p^\star$ as $k \to \infty$.

□

Finally, we note that the proposed OPJLU algorithm adopts the ADMM framework and the difference mainly exists in the automatically updating of two weighting matrices $\boldsymbol{\Omega}_1$ in (22) and $\boldsymbol{\Omega}_2$ in (23).

5. Experiments

In our experiments, we test the performance of the proposed algorithm and compare it with SUnSAL-TV [42], ADSpLRU [31], JSpBLRU [33], BiJSpLRU [32], and MdLRR [41].

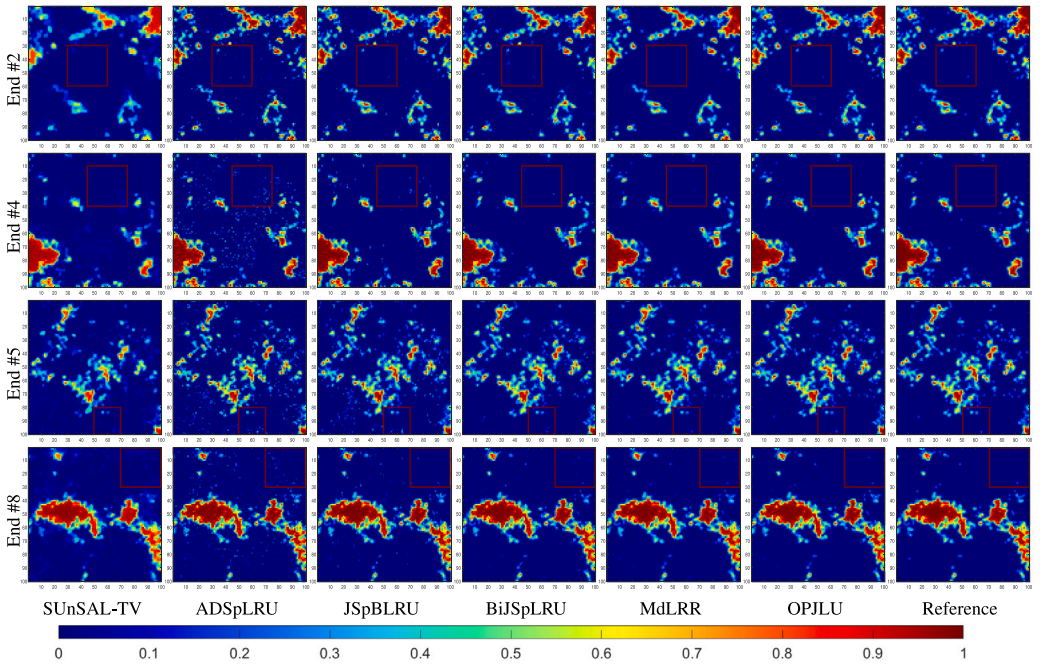


Fig. 4. References and estimated abundance maps by different algorithms on Dataset 1 with SNR = 30 dB.

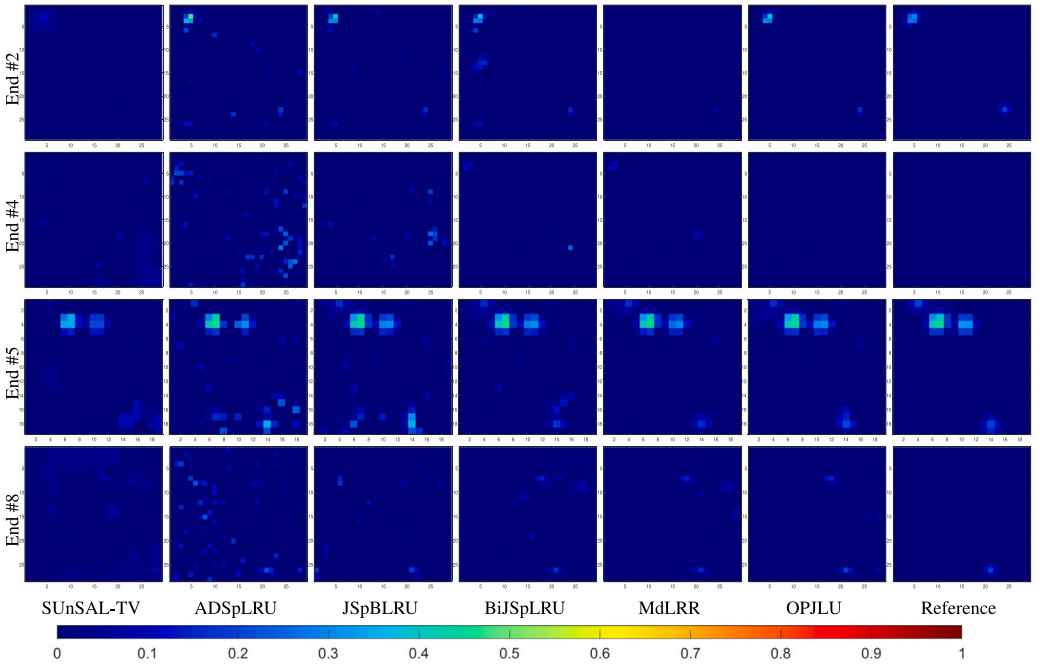


Fig. 5. The detail area in the red boxes in Fig. 4.

Table 2
SRE (dB) and RMSE of different algorithms on Jasper Ridge data and Samson data.

Algorithms		SUnSAL-TV	ADSpLRU	JSpBLRU	BiJSpLRU	MdLRR	OPJLU
Jasper Ridge	SRE	19.76	19.19	20.84	20.72	21.35	**21.83**
data	RMSE	0.0081	0.0086	0.0071	0.0072	0.0067	**0.0064**
Samson	SRE	25.15	24.77	27.25	27.45	27.77	**28.02**
data	RMSE	0.0044	0.0046	0.0034	0.0034	0.0032	**0.0032**

5.1. Experiment setting

All the experiments are implemented using MATLAB (R2024a) on the Windows 10 platform with Intel Core i7-8700k CPU 3.70 GHz and 32 GB of RAM. We use two spectral dictionaries:

- $\boldsymbol{A}_1 \in \mathbb{R}^{224\times240}$ obtains from the U.S. Geological Survey (USGS),[1] which contains 240 materials with 224 spectral bands.
- $\boldsymbol{A}_2 \in \mathbb{R}^{99\times120}$ comes from the National Aeronautics and Space Administration Johnson Space Center Spacecraft Materials Spectral Database. It has 120 materials and 99 bands.

The performance of different algorithms is tested on the following two simulated and two real datasets.

- ***Dataset* 1**: The dataset contains 100×100 pixels, and the corresponding reference abundances are shown in Fig. 3. Randomly select nine spectral signatures from $\boldsymbol{A}_2$ to generate the dataset according to LMM. Then, Gaussian noise is added to the generated dataset with different signal-to-noise ratio (SNR) levels, where
$$\mathrm{SNR} = 10\log_{10}\frac{\|\boldsymbol{Y}\|_F^2}{\|\boldsymbol{E}\|_F^2}, \tag{58}$$
$\boldsymbol{Y}$ represents the generated dataset and $\boldsymbol{E}$ is the added noise. Specifically, Gaussian noise with SNRs of 20 dB, 30 dB, and 40 dB is added, respectively. Generally, a higher value of SNR indicates a lower noise level in the dataset.
- ***Dataset* 2**: This dataset has 128×128 pixels, and its generation method is the same as Dataset 1. Five spectral signatures are selected from $\boldsymbol{A}_1$, and the reference abundance maps are shown in the last column of Fig. 6. We also added noise in the same way as Dataset 1.
- ***Jasper Ridge data***: This data comes from Jasper Ridge in central California, USA and it contains 512×614 pixels and 224 bands. To make it easier to get the real abundance maps, we consider a sub-image with 100×100 pixels and 198 bands. White Gaussian noise with SNR = 20 dB is added to this data. The reference abundance map is shown in the last column of Fig. 7. The spectral library includes 120 spectral curves, four of which are road, soil, water, and tree, see reference spectral curves in Fig. 11 in [46], and 116 spectral curves with 198 bands are randomly selected from $\boldsymbol{A}_1$.
- ***Samson data***: It contains 95×95 pixels with 156 bands. The Samson data assumes that it includes three endmembers, namely soil, tree, and water, with abundances shown in Fig. 8 [53]. It has been added White Gaussian noise with SNR = 20 dB. The spectral library used consists of 120 spectra with 156 bands, generated in the same manner for the Jasper Ridge data.

There are two indexes for testing the performance of all algorithms, which are the signal-to-reconstruction error (SRE) and the root-mean-square error (RMSE). They are defined as

$$\mathrm{SRE\ (dB)} = 10\log_{10}\left(\frac{\sum_{i=1}^n \|\boldsymbol{x}_i\|_2^2}{\sum_{i=1}^n \|\hat{\boldsymbol{x}}_i - \boldsymbol{x}_i\|_2^2}\right), \mathrm{RMSE} = \left(\frac{1}{mn}\sum_{i=1}^n \|\hat{\boldsymbol{x}}_i - \boldsymbol{x}_i\|_2^2\right)^{\frac{1}{2}}, \tag{59}$$

respectively, where $\hat{\boldsymbol{x}}_i$ and $\boldsymbol{x}_i$ are the estimated and reference abundance vectors, respectively. The larger the SRE value and the smaller the RMSE value indicate the better performance of the algorithm. There are three parameters in OPJLU: λ, τ, and μ, which are respectively selected from

$$\{0.0001, 0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5\},$$
$$\{0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1, 5, 10, 50, 100\},$$
$$\{0.001, 0.01, 0.1, 1\}.$$

We take SRE as the indicator and select the parameters of OPJLU by combined cycle parameter adjustment. Recall the primal residual $\boldsymbol{r}^k$ and define the dual residual $\boldsymbol{s}^k$ at the kth iteration as

$$\boldsymbol{r}^k = \boldsymbol{X}^k\boldsymbol{H} - \boldsymbol{V}^k, \quad \boldsymbol{s}^k = \mu(\boldsymbol{V}^k - \boldsymbol{V}^{k-1})\boldsymbol{H}^T. \tag{60}$$

In the proposed algorithm, the iterative stopping conditions are when both

$$\|\boldsymbol{r}^k\|_F \le \eta, \quad \|\boldsymbol{s}^k\|_F \le \eta \tag{61}$$

[1] Available: https://crustal.usgs.gov/speclab/QueryAll07a.php.

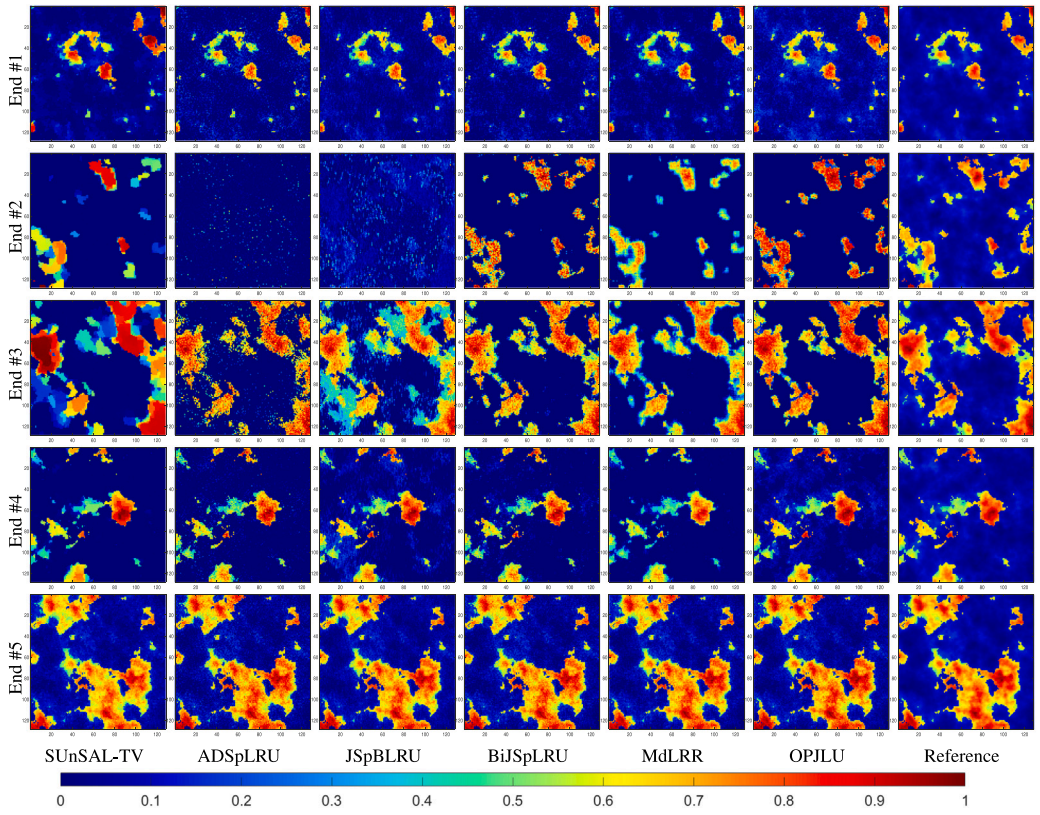


Fig. 6. References and estimated abundance maps by different algorithms on Dataset 2 with SNR = 20 dB.

Table 3
Optimal values of parameters in OPJLU for different datasets.

Parameters	Dataset 1			Dataset 2			Jasper Ridge data	Samson data
	SNR			SNR				
	20 dB	30 dB	40 dB	20 dB	30 dB	40 dB		
λ	0.01	1e−3	1e−4	5e−3	5e−4	1e−4	0.01	0.05
τ	5	5	1	10	10	5	50	100
μ	0.1	0.1	1e-3	1	0.1	1e-3	1	1

are satisfied, or the number of iterations reaches 300. We set $\eta = 5\sqrt{(3m+l)n} \times 10^{-6}$. All the parameter setting methods and the stopping conditions of the comparison algorithms refer to the original articles.

5.2. Experiment results

Table 1 presents the SRE and RMSE results of different algorithms on Datasets 1 and 2 under varying noise levels. From Table 1, we find that OPJLU achieves the highest SREs and the lowest RMSEs in most cases, demonstrating its superiority. Furthermore, compared to BiJSpLRU, the enhanced characterization of sparsity and low-rank properties effectively suppresses noise and improves results. For SNR = 20 dB, where the noise level in Dataset 1 is high, the low-rank property is more effective in suppressing noise, resulting in a better outcome of MdLRR. Moreover, as shown in Table 1, with an increase in SNR, the SRE increases while the RMSE decreases. It can be seen from Fig. 4 that all the algorithms have well performance in the abundance maps obtained on Dataset 1 with SNR = 30 dB. However, OPJLU shows the prominent ability to distinguish noise, so that the obtained abundance maps are more accurate than those obtained by other algorithms. Fig. 5 presents the red areas in Fig. 4. It is clear to display that the abundance maps obtained by OPJLU are more accurate. Fig. 6 shows that the performance of the proposed algorithm in the abundance maps corresponding to endmembers #2, #3, and #5 are close to those obtained by BiJSpLRU. However, the abundance maps corresponding to endmembers #1 and #4 more accurately separate with smaller values on abundance maps, so the results are better than those of other algorithms. From Table 2, we observe that both the SRE and RMSE obtained by OPJLU on the Jasper Ridge and Samson datasets are larger and smaller than those obtained by other algorithms. This shows that OPJLU gets the best results in two real datasets. From Fig. 7, we observe that the abundance maps obtained by all algorithms have satisfactory results.

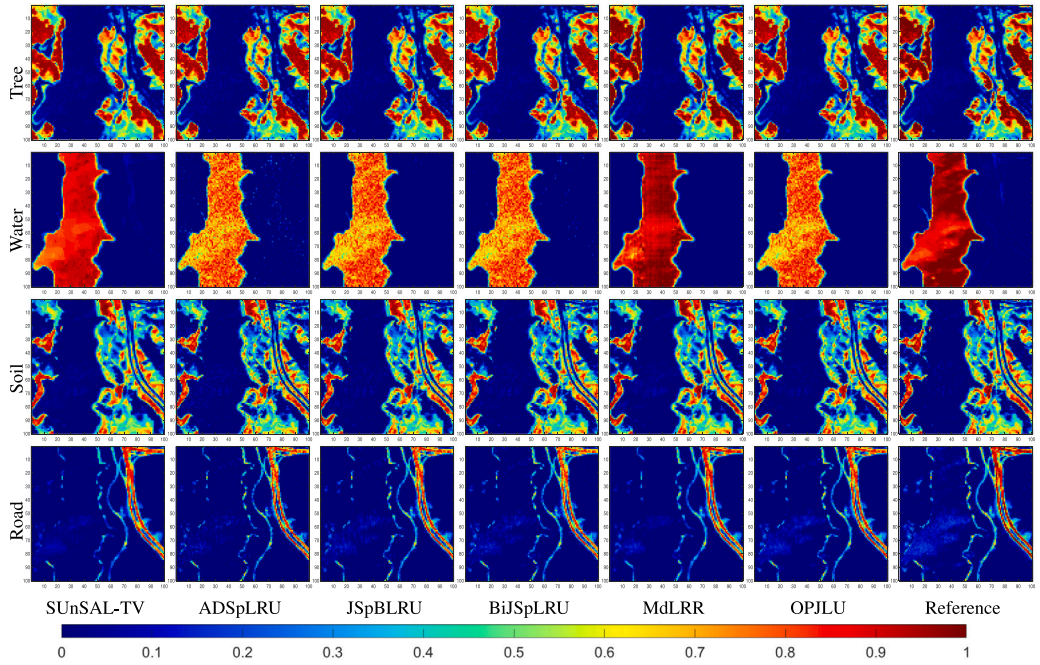


Fig. 7. References and estimated abundance maps by different algorithms on Jasper Ridge data.

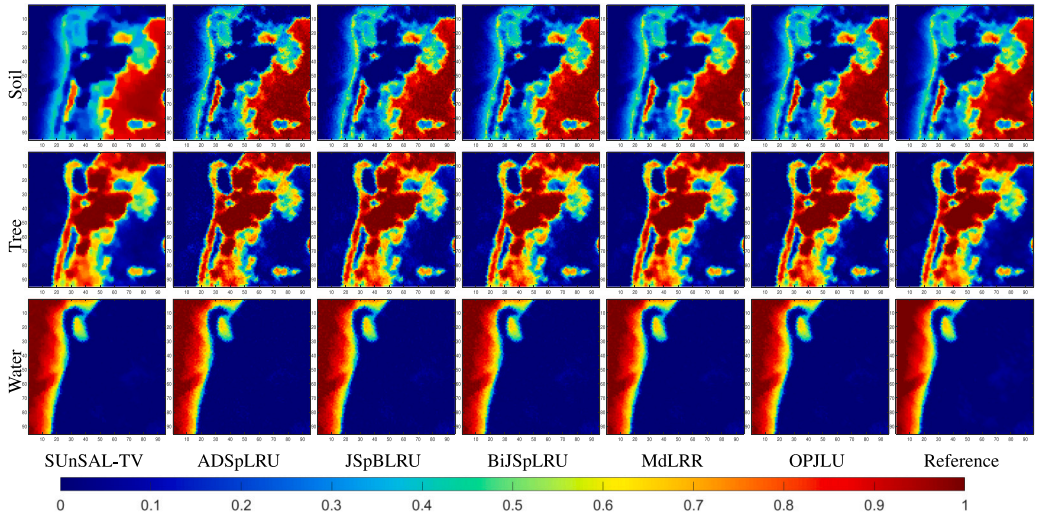


Fig. 8. References and estimated abundance maps by different algorithms on Samson data.

In addition, the abundance map of water obtained by MdLRR is the closest to the reference and the abundance map of soil obtained by our algorithm is more accurate than those obtained by other algorithms. Fig. 8 shows the abundance maps of the Samson data obtained by all the algorithms. From Fig. 8, it can be seen that the results of all methods perform well, but in terms of details, OPJLU is closer to the references.

Fig. 9 illustrates the SREs of OPJLU under parameters λ, τ, and μ on Dataset 1 with different SNRs. It can be observed that as SNR increases, the optimal values of λ, τ, and μ decrease. The optimal values of these parameters for OPJLU in all experiments are shown in Table 3. From Table 3, we see that the optimal values of parameters vary for different datasets and noise levels. Therefore, it is critical to fine tune these parameters in practical cases.

Fig. 10 shows the history of SREs against the number of iterations for different SNRs on Dataset 1. From Fig. 10, we see that the SRE values are stable after 150 iterations, it is sufficient to set the maximum iterations as 300. Additionally, under different noise levels, the convergence of OPJLU tends to be stable as the iteration increases.

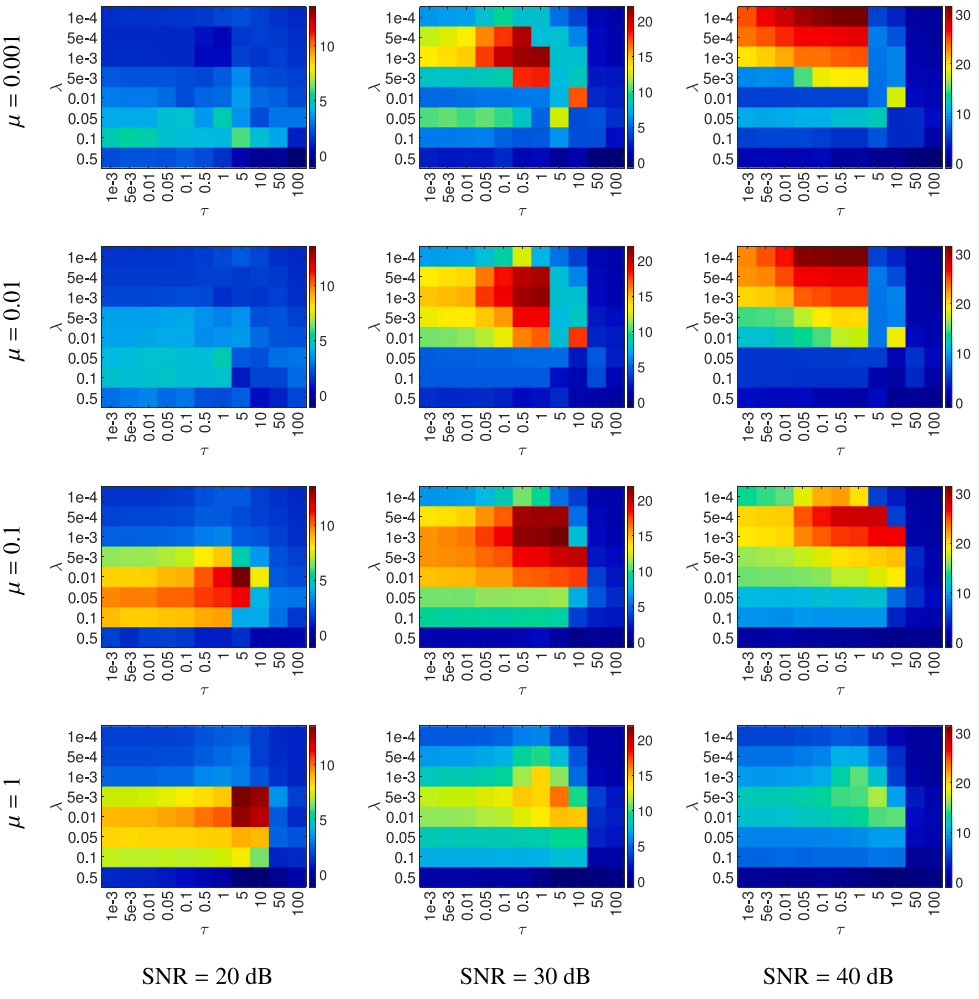


Fig. 9. SRE values versus parameters λ, τ, and μ on Dataset 1 with different SNRs.

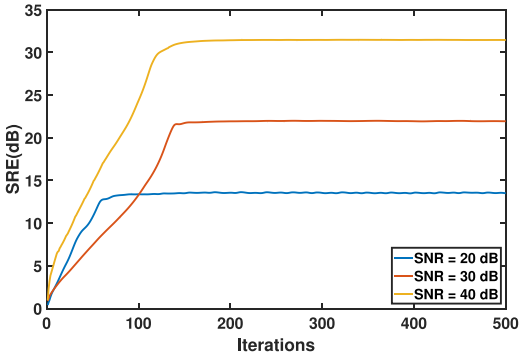


Fig. 10. SRE values versus the iteration numbers on Dataset 1 with different SNRs.

Finally, we give the computational cost of the OPJLU algorithm. Table 4 shows computational time in seconds ($\times 10^{-2}$) of all algorithms after 100 iterations on different datasets. It can be observed from Table 4 that ADSpLRU is the fastest. Since the new block method utilizes each pixel four times, OPJLU is expected to take more time than BiJSpLRU.

Algorithm 2: Pseudocode of the OPJLU-MSD Algorithm.

Input: $\boldsymbol{Y}$, $\boldsymbol{A}$;
Select parameters: λ, τ, μ, ϵ, and maximum iterations.
Initialization: $\tilde{\boldsymbol{X}} = \boldsymbol{0} \in \mathbb{R}^{m\times n}$, $\tilde{\boldsymbol{A}}^0 = \boldsymbol{A}$, $\boldsymbol{V}^0$, $\boldsymbol{\Lambda}^0$, $\tilde{s}^0 = \{1, \dots, m\}$, and set $k = 0$.
repeat
Update $\boldsymbol{X}^{k+\frac{1}{2}} = ((\tilde{\boldsymbol{A}}^k)^T\tilde{\boldsymbol{A}}^k + 6\mu\hat{\boldsymbol{I}})^{-1}((\tilde{\boldsymbol{A}}^k)^T\boldsymbol{Y} + (\mu\boldsymbol{V}^k - \boldsymbol{\Lambda}^k)\boldsymbol{H}^T)$.
Update $\boldsymbol{X}^{k+1}$, $\tilde{\boldsymbol{A}}^{k+1}$, and $\tilde{s}^{k+1}$ by Algorithm 1 in [54].
Update $\Lambda_i^{k+\frac{1}{2}} = \boldsymbol{\Lambda}_i^k(\tilde{s}^{k+1}, :)$, for $i = 1, 2, 3$.
Update $(\boldsymbol{\Omega}_1^{k+1})_{ij} = \dfrac{1}{\|((\boldsymbol{X}^{k+1}\boldsymbol{Q})_j + \frac{1}{\mu}\boldsymbol{\Lambda}_{1,j}^{k+\frac{1}{2}})^{[i]}\|_2 + \epsilon}$ and
$(\boldsymbol{\Omega}_2^{k+1})_{ii} = \dfrac{1}{\sigma_i(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_2^{k+\frac{1}{2}}) + \epsilon}$, for $i = 1, \dots, m$, and $j = 1, \dots, t$.
Update $(\boldsymbol{V}_{1,j}^{k+1})^{[i]} = \textbf{vect-soft}_{\frac{\lambda}{\mu}(\boldsymbol{\Omega}_1^{k+1})_{ij}}(((\boldsymbol{X}^{k+1}\boldsymbol{Q})_j + \frac{1}{\mu}\boldsymbol{\Lambda}_{1,j}^{k+\frac{1}{2}})^{[i]})$, for $i = 1, \dots, m$, and $j = 1, \dots, t$,
$\boldsymbol{V}_2^{k+1} = \textbf{SVT}_{\boldsymbol{\Omega}_2^{k+1}, \frac{\tau}{\mu}}(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_2^{k+\frac{1}{2}})$, and $\boldsymbol{V}_3^{k+1} = \max(\boldsymbol{X}^{k+1} + \frac{1}{\mu}\boldsymbol{\Lambda}_3^{k+\frac{1}{2}}, \boldsymbol{0})$.
Update $\boldsymbol{\Lambda}_1^{k+1} = \boldsymbol{\Lambda}_1^{k+\frac{1}{2}} + \mu(\boldsymbol{X}^{k+1}\boldsymbol{Q} - \boldsymbol{V}_1^{k+1})$, and $\boldsymbol{\Lambda}_i^{k+1} = \boldsymbol{\Lambda}_i^{k+\frac{1}{2}} + \mu(\boldsymbol{X}^{k+1} - \boldsymbol{V}_i^{k+1})$, for $i = 2, 3$.
until some stopping criterion is satisfied;
Update $\tilde{\boldsymbol{X}}(\tilde{s}^{k+1}, :) = \boldsymbol{X}^{k+1}$.
Output: $\hat{\boldsymbol{X}} = \tilde{\boldsymbol{X}}$.

Table 4
Computational time in seconds ($\times 10^{-2}$) of different algorithms on different datasets.

Algorithms	SUnSAL-TV	ADSpLRU	JSpBLRU	BiJSpLRU	MdLRR	OPJLU
Dataset 1	17.19	**5.38**	6.42	10.06	16.74	15.85
Dataset 2	44.64	**22.39**	22.42	32.43	64.26	48.57
Jasper Ridge data	17.93	**6.36**	6.41	10.19	16.76	15.79
Samson data	32.09	**5.77**	6.20	9.60	15.88	15.03

Table 5
SRE (dB) and RMSE of OPJLU, JSpBLRU-MSD, and OPJLU-MSD on different datasets.

Algorithms		Dataset 1			Dataset 2			Jasper Ridge data	Samson data
		SNR			SNR				
		20 dB	30 dB	40 dB	20 dB	30 dB	40 dB		
OPJLU	SRE	13.54	21.99	31.47	12.55	18.59	29.16	21.83	28.02
	RMSE	0.0152	0.0058	0.0019	0.0109	0.0054	0.0016	0.0064	0.0032
JSpBLRU-MSD	SRE	9.98	18.88	28.67	6.85	20.24	29.73	24.04	28.02
	RMSE	0.0230	0.0082	0.0027	0.0210	0.0045	0.0015	0.0049	0.0032
OPJLU-MSD	SRE	**14.03**	**22.02**	**31.52**	**14.80**	**21.01**	**29.98**	**24.80**	**28.74**
	RMSE	**0.0144**	**0.0057**	**0.0019**	**0.0084**	**0.0041**	**0.0015**	**0.0045**	**0.0029**

5.3. Extension

In [37,55], it is considered that in sparse unmixing, only a few endmembers contribute to the formation of mixed pixels, and these endmembers are referred to as active endmembers in [54]. Therefore, a framework named mixed support detection (MSD) is proposed, with a main emphasis on identifying active and nonactive endmembers during each iteration and distinguishing their abundances. This approach aims to lower computational costs, prevent interference from nonactive abundances, and improve the sparsity of the abundances. The MSD framework is embedded into the ADMM updates, leading to an ADMM-MSD algorithm. To consider the detection of active and nonactive endmembers in OPJLU, we introduce the advantages of ADMM-MSD into it. Specifically, OPJLU is also based on the ADMM framework, making it compatible with MSD-ADMM. Thus, we propose a new algorithm, OPJLU-MSD, to further account for the discriminative contributions of active and nonactive endmembers. We showcase the OPJLU-MSD in Algorithm 2.

We present the results of OPJLU, JSpBLRU-MSD, and OPJLU-MSD on all the experimental datasets mentioned above. The ranges of the parameters λ, τ, and μ in OPJLU-MSD are the same as those in OPJLU, $\epsilon = 5e-5$, and maximum iterations are set to 300. The experimental results are shown in Table 5. From Table 5, we find that OPJLU outperforms JSpBLRU-MSD on Dataset 1 while JSpBLRU-MSD outperforms OPJLU on Dataset 2 with SNR being 30 dB and 40 dB and Jasper Ridge data, OPJLU-MSD performs better than JSpBLRU-MSD on all datasets. This experiment further demonstrates the effectiveness of the proposed overlapping patch-based joint-sparse structure.

6. Conclusion

In this article, we have proposed first an overlapping patch-based joint-sparse structure and then an unmixing algorithm OPJLU based on ADMM. We assume that pixels in a patch are composed of the same support set of endmembers. By using this overlapping patch-based joint-sparse structure, more spatial information can be captured, and specifically, each pixel is utilized four times. The residual and objective convergence of OPJLU has been provided in this article. Experiments show that OPJLU is effective for sparse unmixing. Despite that OPJLU improves unmixing performance, it also increases the computational burden. In the future, we will continue to explore more efficient spatial regularization techniques to improve sparse unmixing performance while reducing computational cost. Additionally, researching the combination of deep learning with traditional models like OPJLU is expected to bring improvements by capturing more detailed information in HSIs.

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Data availability

Data will be made available on request.

References

[1] W. He, N. Yokoya, X. Yuan, Fast hyperspectral image recovery of dual-camera compressive hyperspectral imaging via non-iterative subspace-based fusion, IEEE Trans. Image Process. 30 (2021) 7170–7183.
[2] W.-K. Ma, J.M. Bioucas-Dias, T.-H. Chan, N. Gillis, P. Gader, A.J. Plaza, A. Ambikapathi, C.-Y. Chi, A signal processing perspective on hyperspectral unmixing: Insights from remote sensing, IEEE Signal Process. Mag. 31 (1) (2014) 67–81.
[3] J. Peng, W. Sun, H.-C. Li, W. Li, X. Meng, C. Ge, Q. Du, Low-rank and sparse representation for hyperspectral image processing: A review, IEEE Geosci. Remote. Sens. Mag. 10 (1) (2022) 10–43.
[4] X. Shang, M. Song, Y. Wang, C. Yu, H. Yu, F. Li, C.-I. Chang, Target-constrained interference-minimized band selection for hyperspectral target detection, IEEE Trans. Geosci. Remote Sens. 59 (7) (2021) 6044–6064.
[5] D.W.J. Stein, S.G. Beaven, L.E. Hoff, E.M. Winter, A.P. Schaum, A.D. Stocker, Anomaly detection from hyperspectral imagery, IEEE Signal Process. Mag. 19 (1) (2002) 58–69.
[6] H. Zhang, T.-Z. Huang, X.-L. Zhao, W. He, J.K. Choi, Y.-B. Zheng, Hyperspectral image denoising: Reconciling sparse and low-tensor-ring-rank priors in the transformed domain, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–13.
[7] A.F.H. Goetz, G. Vane, J.E. Solomon, B.N. Rock, Imaging spectrometry for earth remote sensing, Science 228 (4704) (1985) 1147–1153.
[8] N. Keshava, J.F. Mustard, Spectral unmixing, IEEE Signal Process. Mag. 19 (1) (2002) 44–57.
[9] J.M.P. Nascimento, J.M.B. Dias, Vertex component analysis: A fast algorithm to unmix hyperspectral data, IEEE Trans. Geosci. Remote Sens. 43 (4) (2005) 898–910.
[10] D.C. Heinz, Chein-I-Chang, Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery, IEEE Trans. Geosci. Remote Sens. 39 (3) (2001) 529–545.
[11] V.P. Pauca, J. Piper, R.J. Plemmons, Nonnegative matrix factorization for spectral data analysis, Linear Algebra Appl. 416 (1) (2006) 29–47.
[12] J.M. Bioucas-Dias, M.A. Figueiredo, Alternating direction algorithms for constrained sparse regression: Application to hyperspectral unmixing, in: 2010 2nd Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, IEEE, 2010, pp. 1–4.
[13] Z. Guo, A. Min, B. Yang, J. Chen, H. Li, J. Gao, A sparse oblique-manifold nonnegative matrix factorization for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–13.
[14] X. Lu, H. Wu, Y. Yuan, P. Yan, X. Li, Manifold regularized sparse NMF for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 51 (5) (2013) 2815–2826.
[15] L. Miao, H. Qi, Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization, IEEE Trans. Geosci. Remote Sens. 45 (3) (2007) 765–777.
[16] Y. Qian, S. Jia, J. Zhou, A. Robles-Kelly, Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization, IEEE Trans. Geosci. Remote Sens. 49 (11) (2011) 4282–4297.
[17] S. Zhang, G. Zhang, F. Li, C. Deng, S. Wang, A. Plaza, J. Li, Spectral-spatial hyperspectral unmixing using nonnegative matrix factorization, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–13.
[18] S. Khoshsokhan, R. Rajabi, H. Zayyani, Sparsity-constrained distributed unmixing of hyperspectral data, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 12 (4) (2019) 1279–1288.
[19] R. Rajabi, H. Ghassemian, Sparsity constrained graph regularized NMF for spectral unmixing of hyperspectral data, J. Indian Soc. Remote. Sens. 43 (2015) 269–278.
[20] S. Khoshsokhan, R. Rajabi, H. Zayyani, Clustered multitask non-negative matrix factorization for spectral unmixing of hyperspectral data, J. Appl. Remote. Sens. 13 (2) (2019) 026509.

[21] R. Rajabi, H. Ghassemian, Spectral unmixing of hyperspectral imagery using multilayer NMF, IEEE Geosci. Remote. Sens. Lett. 12 (1) (2015) 38–42.
[22] Y. Qian, F. Xiong, S. Zeng, J. Zhou, Y.Y. Tang, Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery, IEEE Trans. Geosci. Remote Sens. 55 (3) (2017) 1776–1792.
[23] Y. Fang, Y. Wang, L. Xu, R. Zhuo, A. Wong, D.A. Clausi, BCUN: Bayesian fully convolutional neural network for hyperspectral spectral unmixing, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–14.
[24] P. Ghosh, S.K. Roy, B. Koirala, B. Rasti, P. Scheunders, Hyperspectral unmixing using transformer network, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–16.
[25] B. Rasti, B. Koirala, P. Scheunders, J. Chanussot, MiSiCNet: Minimum simplex convolutional network for deep hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–15.
[26] F. Xiong, J. Zhou, S. Tao, J. Lu, Y. Qian, SNMF-Net: Learning a deep alternating neural network for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–16.
[27] M. Zhao, S. Shi, J. Chen, N. Dobigeon, A 3-D-CNN framework for hyperspectral unmixing with spectral variability, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–14.
[28] M. Zhao, M. Wang, J. Chen, S. Rahardja, Hyperspectral unmixing for additive nonlinear models with a 3-D-CNN autoencoder network, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–15.
[29] M.-D. Iordache, A. Plaza, J. Bioucas-Dias, On the use of spectral libraries to perform sparse unmixing of hyperspectral data, in: 2010 2nd Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, 2010, pp. 1–4, Reykjavik, Iceland.
[30] T. Chen, Y. Liu, Y. Zhang, B. Du, A. Plaza, Superpixel-based collaborative and low-rank regularization for sparse hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–16.
[31] P.V. Giampouras, K.E. Themelis, A.A. Rontogiannis, K.D. Koutroumbas, Simultaneously sparse and low-rank abundance matrix estimation for hyperspectral image unmixing, IEEE Trans. Geosci. Remote Sens. 54 (8) (2016) 4775–4789.
[32] J. Huang, W.-C. Di, J.-J. Wang, J. Lin, T.-Z. Huang, Bilateral joint-sparse regression for hyperspectral unmixing, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 14 (2021) 10147–10161.
[33] J. Huang, T.-Z. Huang, L.-J. Deng, X.-L. Zhao, Joint-sparse-blocks and low-rank representation for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 57 (4) (2019) 2419–2438.
[34] L. Ren, D. Hong, L. Gao, X. Sun, M. Huang, J. Chanussot, Hyperspectral sparse unmixing via nonconvex shrinkage penalties, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–15.
[35] C.Y. Zheng, H. Li, Q. Wang, C.L. Philip Chen, Reweighted sparse regression for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 54 (1) (2016) 479–488.
[36] W. He, H. Zhang, L. Zhang, Sparsity-regularized robust non-negative matrix factorization for hyperspectral unmixing, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 9 (9) (2016) 4267–4279.
[37] M.-D. Iordache, J.M. Bioucas-Dias, A. Plaza, Collaborative sparse regression for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 52 (1) (2014) 341–354.
[38] Q. Qu, N.M. Nasrabadi, T.D. Tran, Abundance estimation for bilinear mixture models via joint sparse and low-rank representation, IEEE Trans. Geosci. Remote Sens. 52 (7) (2014) 4404–4423.
[39] J. Sigurdsson, M.O. Ulfarsson, J.R. Sveinsson, Hyperspectral unmixing with l_q regularization, IEEE Trans. Geosci. Remote Sens. 52 (11) (2014) 6793–6806.
[40] C.G. Tsinos, A.A. Rontogiannis, K. Berberidis, Distributed blind hyperspectral unmixing via joint sparsity and low-rank constrained non-negative matrix factorization, IEEE Trans. Comput. Imaging 3 (2) (2017) 160–174.
[41] L. Wu, J. Huang, M.-S. Guo, Multidimensional low-rank representation for sparse hyperspectral unmixing, IEEE Geosci. Remote. Sens. Lett. 20 (2023) 1–5, Art no. 5502805.
[42] M.-D. Iordache, J.M. Bioucas-Dias, A. Plaza, Total variation spatial regularization for sparse hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 50 (11) (2012) 4484–4502.
[43] S. Boyd, N. Parikh, E. Chu, B. Peleato, J. Eckstein, et al., Distributed optimization and statistical learning via the alternating direction method of multipliers, Found. Trends Mach. Learn. 3 (1) (2011) 1–122.
[44] A. Javaheri, H. Zayyani, M.A.T. Figueiredo, F. Marvasti, Robust sparse recovery in impulsive noise via continuous mixed norm, IEEE Signal Process. Lett. 25 (8) (2018) 1146–1150.
[45] J. Huang, K.-H. Chen, J.-J. Wang, W. Yan, Hyperspectral image denoising via weighted double sparsity total variation and low-rank representation, Inverse Probl. Imaging 18 (5) (2024) 1142–1170.
[46] J.-J. Wang, D.-C. Wang, T.-Z. Huang, J. Huang, X.-L. Zhao, L.-J. Deng, Endmember independence constrained hyperspectral unmixing via nonnegative tensor factorization, Knowl.-Based Syst. 216 (2021) 106657.
[47] P. Yang, T.-Z. Huang, J. Huang, J.-J. Wang, Efficient weighted-adaptive sparse constrained nonnegative tensor factorization for hyperspectral unmixing, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 15 (2022) 10113–10130.
[48] Y.C. Eldar, H. Bolcskei, Block-sparsity: Coherence and efficient recovery, in: IEEE Int. Conf. Acoust., Speech Signal Process, ICASSP, 2009, pp. 2885–2888.
[49] Z. Habibi, H. Zayyani, Markovian adaptive filtering algorithm for block-sparse system identification, IEEE Trans. Circuits Syst. II Express Briefs 68 (8) (2021) 3032–3036.
[50] M. Korki, H. Zayyani, J. Zhang, Bayesian hypothesis testing for block sparse signal recovery, IEEE Commun. Lett. 20 (3) (2016) 494–497.
[51] M. Korki, J. Zhang, C. Zhang, H. Zayyani, Iterative Bayesian reconstruction of non-IID block-sparse signals, IEEE Trans. Signal Process. 64 (13) (2016) 3297–3307.
[52] S.J. Wright, R.D. Nowak, M.A.T. Figueiredo, Sparse reconstruction by separable approximation, IEEE Trans. Signal Process. 57 (7) (2009) 2479–2493.
[53] X. Tao, M.E. Paoletti, Z. Wu, J.M. Haut, P. Ren, A. Plaza, An abundance-guided attention network for hyperspectral unmixing, IEEE Trans. Geosci. Remote Sens. 62 (2024) 1–14, Art no. 5505414.
[54] J. Huang, S. Liang, L.-J. Deng, Mixing support detection-based alternating direction method of multipliers for sparse hyperspectral image unmixing, J. Math. Imaging Vision 66 (6) (2024) 993–1018.
[55] M.-D. Iordache, J.M. Bioucas-Dias, A. Plaza, Sparse unmixing of hyperspectral data, IEEE Trans. Geosci. Remote Sens. 49 (6) (2011) 2014–2039.